Sharp Total Apéry Factorization Multiplicities
and Tie Proliferation along Genuine Farey Parent–Child Steps

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Abstract

For a primitive positive signed three-letter source $(p, -q, -s)$, we count the number of additional minimum-length original-labelled Parikh factorizations across its p fundamental residual classes. We identify this statistic exactly with the total excess number of factorizations of the Apéry set of the numerical semigroup $H = \langle p, p+q, p+s\rangle$. When H is minimally three-generated, its sharp maximum at fixed multiplicity p is

$$M_3(p) = \begin{cases} \lceil (p-3)^2/12 \rceil, & p \text{ odd}, \\ \lfloor (p-3)^2/12 \rfloor, & p \text{ even}. \end{cases}$$

For $p = 5$ and all $p \geq 7$, equality is characterized precisely by the coprime-reduced pair of nontrivial generating costs being $(2, 3)$. We then construct, for *every* $p \geq 7$, a strictly positive unimodular rank-three Farey parent–child step at the bound, despite both original parent optimal words being entirely tie-free. The projected parent determinant and joint cost determinant are unbounded; one parent's word also lacks the usual binary optimum shadow. On this family an explicit bounded-extension reblocking of an actual optimized parent word produces every daughter optimum and tie without a daughter min-plus minimization. We separate these restricted constructions from the still-open unrestricted original-parent-word transfer question and from classical general numerical-semigroup factorization algorithms.

Keywords. numerical semigroup; Apéry set; factorization multiplicity; Farey parent word; optimal Parikh fiber; L-shaped minimum-distance diagram.

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1 Introduction and relation to previous work

Let p, q, s be positive integers with $\gcd(p, q, s) = 1$. For each $0 \leq r < p$, we study all nonnegative integral solutions of

$$px - qu - sv = r \tag{1}$$

that minimize the *literal* length $x + u + v$. Keeping u and v distinct is essential, even if the two negative letters have equal magnitude. In word applications these solutions are original-labelled Parikh records, not interchangeable unlabelled paths.

The embedding of (1) into a numerical semigroup is elementary, and general algorithms for three-generator numerical-semigroup factorizations and L-shaped weighted double-loop diagrams already exist; see [1, 2]. Moreover, Autry, Gomes, O'Neill and Ponomarenko [3] explicitly study factorization nonuniqueness within Apéry sets: their Theorem 4 already uses three-generated semigroups with coprime-reduced nontrivial generators $(2, 3)$ and constructs families with many non-half-factorial Apéry elements. Thus neither the L-shape method, the

$(2,3)$ generator pattern, nor its ability to produce many nonunique Apéry factorizations is claimed here as new. Our narrower question counts *all additional factorizations with multiplicity*, sums them over the Apéry set at fixed smallest generator p, and asks for its exact sharp global maximum and Farey parent–child realizability.

Contributions and novelty

The main contributions of this article are the following. First, we give the exact sharp total excess multiplicity for minimally three-generated semigroups at each fixed p, together with an equality characterization. Second, we construct an authentic positive $\mathrm{SL}_3(\mathbb{Z})$ Farey edge for *every* $p \geq 7$ at equality, whose two fully optimized original parent words contain no ties. Third, we give a restricted word reblocking producing the complete daughter optimum fibers from an already optimized original parent word. This is *not* an unrestricted order-only Farey word recursion; the family is special, and its parental input is reconstructible from source coefficients.

We distinguish proved mathematical statements from historical priority. The cited 2010 paper gives complete factorization algorithms, and the 2007 paper studies monomial ideals and minimum-distance diagrams; these precedents must not be represented as novel components here. The exact fixed-p global *total excess factorization count* extremum and the joint authentic-Farey realization have not been located verbatim in the sources checked for this draft; this does not establish historical firstness. In particular, the already-known $(2,3)$ Apéry extremizing examples in [3] overlap the semigroup family here, but the statistics addressed there concern the proportion and lengths of nonunique factorizations rather than the fixed-p sum of their *numbers*.

2 Minimum Parikh fibers and the Apéry correspondence

Write $\mathbb{N} = \mathbb{Z}_{\geq 0}$. Let

$$\mathcal{M}_{p,q,s}(r) = \operatorname*{arg\,min}_{\substack{(x,u,v)\in\mathbb{N}^3 \\ px-qu-sv=r}} (x+u+v), \qquad E(p,q,s) = \sum_{r=0}^{p-1} (|\mathcal{M}_{p,q,s}(r)| - 1).$$

All these sets are nonempty, since $\gcd(p,q,s) = 1$ and the two negative letters generate all residues modulo p. Define $a = p+q$, $b = p+s$ and the numerical semigroup $H = \langle p, a, b\rangle$. Since $a, b > p$, its multiplicity is p.

Proposition 2.1 (Apéry–Parikh bijection)**.** *For every $r \in \{0,\dots,p-1\}$ let w_r be the least element of H congruent to $-r$ modulo p. Then $\mathcal{M}_{p,q,s}(r)$ is in bijection with the nonnegative (a,b)-factorizations of w_r. Consequently, if $d_H(w)$ is the number of factorizations of w using the minimal generating set when* $\operatorname{edim}(H) = 3$*, then*

$$E(p,q,s) = \sum_{w\in \mathrm{Ap}(H,p)} (d_H(w) - 1). \tag{2}$$

Proof. The source equation gives the exact identity

$$p(x+u+v) = r + (p+q)u + (p+s)v = r + au + bv.$$

Conversely, if $au + bv \equiv -r \pmod p$ and $u, v \geq 0$, then $x = (r + qu + sv)/p$ is a nonnegative integer and yields (1). Thus minimizing length is exactly minimizing $au + bv$ in its congruence class. A factorization of an Apéry element cannot use p, since subtracting p would remain inside H. The claimed correspondence and (2) follow. □

Remark 2.2. *If $q = s$, primitivity forces $\gcd(p, q) = 1$. The unique smallest total negative count k_r ranges over $0, \dots, p-1$ as r varies, and it has $k_r + 1$ physically distinct splits into (u, v). Hence $E = p(p-1)/2$ in the two-effective-generator symmetric case. The rest of the main extremal theorem concerns genuine embedding dimension three.*

3 The exact two-cost tie count

We give an elementary reduction establishing the exact total count that underlies the extremal theorem. It is a two-dimensional L-shape argument, not a new general factorization algorithm.

After exchanging the negative labels if necessary, write

$$a = gB, \qquad b = gA, \qquad 1 \le A \le B, \quad \gcd(A, B) = 1,$$

where $g = \gcd(a, b)$, so the effective weight of (u, v) is $g(Bu + Av)$. The case $A = B = 1$ has already been addressed. For $A < B$, choose the unique integer $h \in \{1, \dots, A\}$ satisfying

$$Bh \equiv p \pmod{A}, \qquad t = \frac{p - Bh}{A} \in \mathbb{Z}. \tag{3}$$

Lemma 3.1 (Exact positive-coin tie mass)**.** *With this notation, if $t \le 0$ then $E = 0$. If $t \ge 1$, then*

$$p = At + Bh, \qquad E = ht + A\sum_{j \ge 1}(t - jB)_+, \quad (z)_+ = \max(z, 0). \tag{4}$$

Moreover, in the positive-t case a canonical set of one cost-minimizing representative per residue is the L-shaped set

$$\mathcal{D} = \big([0, h) \times [0, B + t)\big) \cap \mathbb{Z}^2 \ \cup \ \big([h, A) \times [0, t)\big) \cap \mathbb{Z}^2. \tag{5}$$

Every canonical $(u, v) \in \mathcal{D}$ has exactly $1 + \lfloor v/B \rfloor$ tied nonnegative minimum-cost representatives.

Proof. Assume $t \ge 1$. The set (5) contains $h(B + t) + (A - h)t = Bh + At = p$ points. Starting with any nonnegative pair (u, v), apply the following moves, preserving the residue of $g(Bu + Av)$ modulo p:

$$\begin{aligned}
&(u, v) \longmapsto (u - A, v + B) && (u \ge A),\\
&(u, v) \longmapsto (u - h, v - t) && (h \le u < A,\ v \ge t),\\
&(u, v) \longmapsto (u + A - h, v - B - t) && (u < h,\ v \ge B + t).
\end{aligned}$$

The first move preserves weight and reduces u; each other move decreases weight by exactly gp, hence strictly. Iterating terminates with a point of $\mathcal{D}$, and every residue is represented, since $\gcd(p, a, b) = 1$. As $\mathcal{D}$ has exactly p points, its residue representatives are distinct. Since an arbitrary competing pair reduces to a point of $\mathcal{D}$ without an increase in weight, these representatives are globally minimum-cost.

Two nonnegative pairs of the same cost differ by an integer multiple of $(A, -B)$. As a canonical pair has $0 \le u < A$, all other nonnegative representations of that cost are precisely $(u + jA, v - jB)$ for $0 \le j \le \lfloor v/B \rfloor$. Therefore

$$E = h\sum_{v=0}^{B+t-1}\left\lfloor \frac{v}{B} \right\rfloor + (A - h)\sum_{v=0}^{t-1}\left\lfloor \frac{v}{B} \right\rfloor = ht + A\sum_{j \ge 1}(t - jB)_+.$$

Now assume $t = -k \le 0$. A tied Apéry factorization would have two pairs differing by $(A, -B)$, and one of them has $u \ge A \ge h$. If its weight is $w = g(Bu + Av)$, then

$$w - gp = g\big(B(u - h) + A(v + k)\big) \in \langle a, b \rangle.$$

Thus $w - p = (w - gp) + (g - 1)p \in H$, contradicting the Apéry property. No ties exist. □

Corollary 3.2 (Tie segments)**.** *For any certified optimum record $z = (x, u, v)$ in a fixed residual fiber, all tied optimum records are exactly the nonnegative integer points in*

$$z + \mathbb{Z}\,\frac{(q - s,\ p + s,\ -(p + q))}{\gcd(p + q, p + s)}. \tag{6}$$

They form one consecutive finite interval in the integer parameter.

Proof. Two optimum triples have the same residual and the same literal length. Their difference solves two independent homogeneous integral equations, whose primitive rank-one kernel is the displayed vector. Conversely each nonnegative point along it preserves residual and length. □

4 Sharp total multiplicity at fixed embedding dimension

Theorem 4.1 (Exact three-generator maximum)**.** *For every integer $p \geq 3$, among all primitive positive sources (p, q, s) for which $H = \langle p, p + q, p + s\rangle$ is minimally three-generated, the largest possible total extra optimum mass is*

$$\boxed{M_3(p) = \begin{cases} \left\lceil \frac{(p-3)^2}{12} \right\rceil, & p \text{ odd}, \\ \left\lfloor \frac{(p-3)^2}{12} \right\rfloor, & p \text{ even}. \end{cases}} \tag{7}$$

For $p = 5$ or $p \geq 7$, equality holds if and only if the unordered coprime-reduced pair of nontrivial costs is $\{2, 3\}$:

$$\left\{ \frac{p + q}{g}, \frac{p + s}{g} \right\} = \{2, 3\}, \qquad g = \gcd(p + q, p + s). \tag{8}$$

The exceptional moduli $p = 3, 4, 6$ have $M_3(p) = 0$ and nonunique equality descriptions. Every such three-generated source satisfies $E(p, q, s) < p^2/12$, and the constant $1/12$ is asymptotically best possible.

Proof. Minimal generation forces $A \geq 2$: if $A = 1$, one nontrivial generator divides the other. With $A < B$ and $\gcd(A, B) = 1$, this gives $B \geq 3$. If $t \leq 0$, Lemma 3.1 gives $E = 0$. On the tie-active branch $t \geq 1$, using the strict comparison of a positive arithmetic sum with the area below its piecewise-linear interpolation yields

$$\sum_{j \geq 1} (t - jB)_+ < \frac{t^2}{2B}.$$

By (4), $h = (p - At)/B$, whence

$$E < ht + \frac{At^2}{2B} = \frac{pt - At^2/2}{B} \leq \frac{p^2}{2AB} \leq \frac{p^2}{12}. \tag{9}$$

For $(A, B) = (2, 3)$, the canonical h is 1 when p is odd and 2 when p is even, and $t = (p - 3h)/2$. When $t > 0$, write $t = 3k + \rho$, $0 \leq \rho < 3$. The exact sum in (4) becomes

$$E = h(3k + \rho) + 3k(k - 1) + 2k\rho. \tag{10}$$

Elementary division by 6 in (10) gives the two cases in (7); when $t \leq 0$ the same formula gives the stated zero maxima for the small exceptional cases. The pair $(2, 3)$ is realized by $a = 2(p+1)$, $b = 3(p + 1)$ for every $p \geq 5$: $\gcd(p, p + 1) = 1$, and $\langle p, a, b\rangle$ is minimally three-generated. For completeness, the exceptional small multiplicities $p = 3, 4$ also admit minimally three-generated examples with zero excess mass, namely $\langle 3, 4, 5\rangle$ and $\langle 4, 5, 6\rangle$; for $p = 6$ one may take $\langle 6, 14, 21\rangle$. Thus the values in (7) are attained.

Suppose A, B is any other coprime pair with $2 \le A < B$. If $A = 2$, then $B \ge 5$; otherwise $A \ge 3$ and $B \ge 4$. In either case $AB \ge 10$, so (9) gives

$$E < \frac{p^2}{20}. \tag{11}$$

For $p \ge 16$ one checks directly that

$$M_3(p) \ge \frac{(p-3)^2}{12} - 1 > \frac{p^2}{20}.$$

For $p = 13, 14, 15$ the values of M_3 are $9, 10, 12$, respectively, each strictly greater than $p^2/20$. For $3 \le p \le 12$, the only positive-tie pairs obey $p = At + Bh$ with $2 \le A < B$, $\gcd(A, B) = 1$, $1 \le h \le A$ and $t \ge 1$, so there are finitely many possibilities. The exact check is recorded in Table 1; it uses only (4) and the indicated integer constraints. Hence outside the zero-maximum exceptional cases, only $(A, B) = (2, 3)$ can attain (7). □

p	3	4	5	6	7	8	9	10	11	12
$M_3(p)$	0	0	1	0	2	2	3	4	6	6
Best other (A, B)	0	0	0	0	1	1	2	2	3	2

Table 1: Finite verification of the equality classification at small multiplicity. A dash-free zero denotes no positive-tie competitor.

5 An authentic extremal Farey edge at every multiplicity

A genuine directed Farey edge here means: $F \in \mathrm{SL}_3(\mathbb{Z})$ has strictly positive entries, $P = F(1, 1, 1)^{\mathsf{T}}$ is the complete row-sum source, $Q = Fe_1$ is its true inserted-column source, and $C = P + Q$ is the child produced by the legal first-column-to-second-column addition. All source components are positive and every optimum is in the original three physical labels.

Theorem 5.1 (Tie-free parents and maximally tied children)**.** *For every integer $n \ge 0$, put*

$$F_n = \begin{pmatrix} 2 & n+2 & 1 \\ 1 & n+2 & 1 \\ 3 & 2n+3 & 2 \end{pmatrix} \in \mathrm{SL}_3(\mathbb{Z}). \tag{12}$$

The authentic parents and child are

$$P_n = (n+5, n+4, 2n+8), \quad Q_n = (2, 1, 3), \quad C_n = (p, p-2, 2p-3),\ p = n+7. \tag{13}$$

Both complete original parent optimum words have one record at each fundamental target, whereas

$$E(C_n) = M_3(p) = \Theta(p^2).$$

The projected binary-parent determinant equals $p - 4$ and the joint parent cost determinant equals $p - 1$. Thus both are positive and unbounded, and the first parent has no $v = 0$ optimum record at target 2.

Proof. Subtracting the second row of (12) from the first gives $(1, 0, 0)$; the resulting bottom-right minor is $2(n + 2) - (2n + 3) = 1$. This proves positivity and determinant one. Summing rows and selecting the first column gives (13). The child effective costs are

$$a_C = p + (p-2) = 2(p-1), \qquad b_C = p + (2p-3) = 3(p-1).$$

Because $\gcd(p, p-1) = 1$ and $p \geq 7$, the three numbers $p, 2(p-1), 3(p-1)$ form a minimal generating set: the two larger generators are not multiples of p, and an expression of $3(p-1)$ as a nonnegative combination of p and $2(p-1)$ would require $p \mid 1$ or $p \mid 3$, an impossibility. Hence $\operatorname{edim}(H) = 3$, its reduced cost pair is $(2, 3)$, and Theorem 4.1 gives the asserted equality.

For the first parent set $h = n + 5$; then $P = (h, h-1, 2(h-1))$. Put $w = u + 2v$. The residual equation $hx - (h-1)w = r$ forces $w \equiv r \pmod{h}$, so $w = r + kh$ with $k \geq 0$ and $x = r + k(h-1)$. For fixed w, the unique shortest split is

$$u = w \bmod 2, \qquad v = \lfloor w/2 \rfloor.$$

Its length strictly increases with k. Consequently the entire actual first-parent optimum word is

$$\mathcal{M}_P(r) = \left\{ \left(r, r \bmod 2, \lfloor r/2 \rfloor\right) \right\}, \qquad 0 \leq r < h. \tag{14}$$

In particular $\mathcal{M}_P(2) = \{(2, 0, 1)\}$, so this parent lacks a binary $v = 0$ optimum shadow. For $Q = (2, 1, 3)$ direct inspection gives

$$\mathcal{M}_Q(0) = \{(0, 0, 0)\}, \qquad \mathcal{M}_Q(1) = \{(1, 1, 0)\}.$$

Both parents therefore have exactly one shortest original-labelled record per target. The projected determinant and price determinant calculate as

$$\begin{aligned} 2(n+4) - (n+5) &= n + 3 = p - 4, \\ (2n+9) \cdot 5 - 3(3n+13) &= n + 6 = p - 1. \end{aligned}$$

□

6 Complete physical daughter word and a restricted parent reblocking

Theorem 6.1 (Exact full child fibers)**.** *For $C = (p, p-2, 2p-3)$, $p \geq 7$, set*

$$t_r = \begin{cases} 0, & r = 0, \\ p + 1, & r = 1, \\ r, & 2 \leq r < p. \end{cases} \tag{15}$$

Then for each $0 \leq r < p$, the entire *set of original-labelled optimum records, including all ties, is*

$$\boxed{\mathcal{M}_C(r) = \left\{ \left(u + 2v - \frac{t_r - r}{p}, u, v \right) : u, v \in \mathbb{N},\ 2u + 3v = t_r \right\}.} \tag{16}$$

Equivalently, writing

$$\varepsilon_r = t_r \bmod 2, \quad k_r = \lfloor t_r/2 \rfloor, \quad u_r^0 = k_r - \varepsilon_r, \quad x_r^0 = k_r + \varepsilon_r - \frac{t_r - r}{p},$$

we have the compressed arithmetic-progression description

$$\mathcal{M}_C(r) = \left\{ (x_r^0 + j, u_r^0 - 3j, \varepsilon_r + 2j) : 0 \leq j \leq \left\lfloor \frac{u_r^0}{3} \right\rfloor \right\}. \tag{17}$$

The optimum length profile is $L_C(0) = 0$, $L_C(1) = p$, $L_C(r) = r$ for $2 \leq r < p$.

Proof. The child residual decomposes as

$$px - (p-2)u - (2p-3)v = p(x-u-2v) + (2u+3v).$$

For a feasible triple let $t = 2u+3v$. Then $t \equiv r \pmod{p}$ and

$$x = u + 2v - \frac{t-r}{p}, \qquad x+u+v = \frac{r+(p-1)t}{p}.$$

All nonnegative integers except 1 are representable by 2 and 3. Thus the least feasible t in each residue is exactly (15); its indicated x is nonnegative. Any larger feasible t strictly raises the length. The full optimum set is therefore (16). Solutions of $2u+3v = t_r$ have $v = \varepsilon_r + 2j$ and $u = k_r - \varepsilon_r - 3j$, proving (17) and the length formula. □

Corollary 6.2 (A restricted actual-parent-word transfer)**.** *Let $Z_P(t) = (t, \varepsilon_t, k_t)$ denote the consecutive* true original optimized *first-parent marks in* (14)*, where $\varepsilon_t = t \bmod 2$ and $k_t = \lfloor t/2 \rfloor$. Their successive literal increments alternate between $(1,1,0)$ and $(1,-1,1)$. Extend this alternating word at most four places past its last parent index $h-1 = p-3$, so that the mark at $t_r \le p+1$ is available. If the resulting mark is (t_r, ε, k), put*

$$z_C^0(r) = \left(k + \varepsilon - \frac{t_r - r}{p},\ k - \varepsilon, \varepsilon\right).$$

Then the complete child fiber follows without a daughter min-plus minimization *as*

$$\mathcal{M}_C(r) = \left\{ z_C^0(r) + j(1,-3,2) : 0 \le j \le \left\lfloor \frac{k-\varepsilon}{3} \right\rfloor \right\}.$$

The compact form of the full child word is computed with $O(p)$ integer-arithmetic operations, independent of the number of additional tied optima. Explicitly writing all records requires $\Theta(p + E(C))$ output operations.

Proof. The input parent word supplies $k = \lfloor t_r/2 \rfloor$ and $\varepsilon = t_r \bmod 2$; the alternating increments determine the at-most-four required extensions uniquely. Substitution into (17) proves each coordinate of the asserted reblocking, including its nonnegativity and all tie endpoints. This is a direct source-to-child construction on the displayed family, with no optimization step for the daughter. □

Remark 6.3 (Limit of the transfer claim)**.** *The two parent words are genuine, independently optimized original-labelled inputs, and the transfer operates on their observed ordered symbols. However, the same alternating pattern can be reconstructed from the explicit source coefficients. Moreover, the second parent is not informationally necessary in this particular family. Hence Corollary 6.2 proves neither information-theoretically essential dependence on* both *parental words nor an unrestricted genuine two-parent optimization-free transfer theorem. The broader question is stated below.*

Open problem and future work

Open problem (unrestricted original-parent-WORD transfer). For an arbitrary genuine, strictly positive unimodular rank-three Farey parent–child step, can the entire original-labelled minimum-length daughter WORD, including every optimal tie, be recovered from the two independently optimized, complete ordered parent WORDs and their source-frame data by a general transfer procedure that does not perform a new daughter minimization? The family-specific bounded-extension reblocking proved in Corollary 6.2 is not such a general theorem; it also does not establish that both parents are informationally necessary.

7 Verification and scope of the publication claim

All proofs above are symbolic; the reported finite exact-arithmetic checks serve as independent consistency tests, not substitutes for universal proofs. The checks compared complete optimal original-labelled Parikh fibers, including tied records, against the stated formulas using separate nonnegative weighted-residue shortest-path calculations. The reported verification ranges comprise normalized reduced-cost scans for $3 \leq p \leq 200$, more than 25,000 primitive minimally three-generated direct enumerations, constructive equality checks through $p = 600$, and 150 strictly positive unimodular Farey edges F_n with their complete original parent and child words. These are the stated finite ranges, not an assertion of an exhaustive computational proof; the $p = 3, 4, 6$ exceptions to the equality characterization remain explicit in the theorem.

As to priority, the Apéry description, L-shape basis, and single-element denumerant algorithms are established in earlier literature [1, 2]; the $(2, 3)$ nonunique-Apéry construction also appears in [3, Theorem 4]. Those precursors are treated here as context, not new contributions. What remains to be evaluated by wider specialist literature review is the exact fixed-multiplicity *sum* extremum (7), its unique $(2, 3)$ equality characterization, and its genuine positive Farey realization with tie-free parents. An absence of a matching formula in the two directly reviewed papers is not a certificate of historical originality. This is a self-contained *research manuscript*, not a record of journal submission or acceptance; a definitive global literature-priority assessment remains outside the scope of the paper.

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References

[1] F. Aguiló-Gost and P. A. García-Sánchez, *Factoring in embedding dimension three numerical semigroups*, Electronic Journal of Combinatorics **17** (2010), R138. doi:10.37236/410.

[2] D. Gómez, J. Gutierrez and À. Ibeas, *Cayley digraphs of finite abelian groups and monomial ideals*, SIAM Journal on Discrete Mathematics **21** (2007), 763–784. doi:10.1137/050646056.

[3] J. Autry, T. Gomes, C. O'Neill and V. Ponomarenko, *Elasticity in Apéry sets*, The American Mathematical Monthly **127** (2020), 744–749. doi:10.1080/00029890.2020.1790926.